

Moritz Hauer

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Summary

Software engineer with around five years of experience across C++, Python and TypeScript, from simulation frameworks for industrial systems to full-stack web applications shipped and deployed end to end. Comfortable owning a product across architecture, tests, CI and deployment.

Education

Karlsruhe Institute of Technology (KIT)

M.Sc. Computer Science

Apr 2018 – Oct 2021

Karlsruhe, Germany

- Focus on machine learning, robotics, and embedded systems. Master's thesis: *Indoor Sound Source Localization with Multilateration*.

Offenburg University of Applied Sciences

B.Sc. Applied Computer Science

Apr 2014 – Mar 2018

Offenburg, Germany

- Bachelor's thesis: *Development of a software updater for an injection moulding machine running VxWorks*.

Experience

ARBURG GmbH + Co KG

Software Developer, Digital Twin

Oct 2022 – Present

Karlsruhe, Germany

- Designed the architecture of DTSim, a digital twin that simulates every physical input of an injection moulding machine (axes, drives, sensors, I/O), so the unmodified control software runs on a virtual machine and produces real cycle times and signals [C++, Python].
- Built it as a plugin framework with multiple fidelity levels, letting domain teams contribute simulation components through defined interfaces; two-person core team, used daily by ~50 developers.
- Lets control software be developed and tested before the physical machine exists, with nightly automated tests on all branches; now rolling out to sales and customers, who program their processes on a virtual machine before delivery.
- Member of the software architecture team since Jul 2026.

INERATEC GmbH

Software Engineer, Automation

Oct 2021 – Sep 2022

Karlsruhe, Germany

- Developed a digital twin for a regenerative energy production site to support simulation, testing, and remote monitoring before deployment.
- Implemented data processing and monitoring tools [Python with FastAPI & pandas, REST APIs via Node.js, PostgreSQL, React frontend]

Chair of Dependable Nano Computing, KIT

Research Assistant

Dec 2019 – Jul 2020

Karlsruhe, Germany

- Reproduced and extended side-channel and fault-injection attacks on FPGA platforms [VHDL].
- Improved experimental setups and Python analysis tools for studying reliability and security properties.

ARBURG GmbH + Co KG

Software Engineer, Test Automation (Part-time / Bachelor Thesis)

Sep 2016 – Feb 2018

Loßburg, Germany

- Developed automated test systems for industrial machine controllers, including test case design [C#, C++].

Projects

BetterBeaver | TypeScript, React, Supabase/PostgreSQL, Vitest, GitHub Actions | betterbeaver.de

- Designed, built and deployed a full-stack learning platform single-handedly: React progressive web app with offline support, Supabase/PostgreSQL backend, pnpm monorepo, around 200 commits.
- Implemented user accounts and data access on a Supabase/PostgreSQL backend, automated tests with Vitest, and continuous integration and deployment via GitHub Actions.
- Wrote and maintained in-repository architecture and requirements documentation to keep design decisions reviewable.

Skills

Programming languages: TypeScript, Python, C++, C

Web and data: React, Progressive Web Apps, PostgreSQL, Supabase, FastAPI, Node.js, REST APIs, Vite

Engineering practices: Automated testing (Vitest, pytest), CI/CD with GitHub Actions, code review, architecture documentation

Tooling: Git, Linux, Docker, AI-assisted development (coding agents: GitHub Copilot, Claude Code)

Certificates: ISTQB Certified Tester - Foundation Level, IDTA Digital Twin Implementation Training

Languages: German (native), English (fluent), Kyrgyz (basic)

Community: Karlsruhe Natural Science Association, Makerspace FabLab Karlsruhe, Foodsharing Karlsruhe